

80% precision was a threshold away, and the model was never the problem

Results

Three arms, one evaluator, the eight sealed data/2-eval corpora. **Nothing was retrained**: the two new arms differ from the incumbent in 61 float comparison points and in nothing else.

	baseline scorecard61	p80r70	p80r90
micro F1 (<i>the ranker</i>)	0.7299	0.8394	0.8474
micro precision	0.6360	0.9165	0.8844
macro F0.5	0.6230	0.7093	0.7056
macro F2	0.6651	0.6406	0.6564
precision, macro over catalogue	0.6244	0.7616	0.7456
recall, macro over catalogue	0.6773	0.5668	0.5872
worst measurable priority recall	0.6267	0.4830	0.4921
prediction rate	0.8181	0.7656	0.7855
document recall / precision	0.7975 / 0.8956	<i>identical</i>	<i>identical</i>
one-core p95	4.029 ms	4.029 ms	4.029 ms

Document-level metrics are identical across all three because the gate is upstream of every tag head and no gate parameter moved.

Per-tag precision ≥ 0.80 , on the sealed corpora

arm	pairs passing	of measurable
baseline	102	210
p80r70	141	180
p80r90	139	181

The denominator moves because a model that fires less has fewer tag $\times$ corpus pairs with 30+ predictions, and a pair with too few predictions is NOT_MEASURABLE rather than failed. So the honest comparison is the **absolute** count: 141 passes against 102, an extra 39 tag $\times$ corpus pairs conclusively above 80% precision.

Where the remaining failures are

p80r70, per-tag precision:

corpus	pass	fail
20000_pii_holdout	47	2
38937_openpii	27	1
30000_pii2_eval	46	7
10626_ai4privacy	20	18
10360_betterdataai	1	11

29 of the 39 failures are on the two corpora with known labelling defects. On the other three, 120 of 130 measurable pairs clear 80% precision. betterdataai is silver-labelled — its labels are themselves model output — and 85.5% of its name-bearing rows carry `full_name` alone, which is why `middle_name` there tops out near precision 0.004 at *any* threshold. That is a gold ceiling, not a model ceiling, and it was measured before this run started.

Recall ≥ 0.70 passes 44 of 55 measurable pairs; 3 of the 11 failures are on `datax` and `govdocs2`, which carry positive-only tag gold and can measure nothing else.

Both arms are formally blocked

NO FEASIBLE ARM under both policies. "Across the board" means every measurable pair, and 141 of 180 is not 180 of 180. Nothing was promoted.

TL;DR

- **The target was reachable on 54 of 61 tags by threshold alone** — measured on the calibration curve *before* anything was built, which is what separated "needs a threshold" from "needs a model".
- **Micro F1 0.7299 → 0.8394 and micro precision 0.6360 → 0.9165**, with no retraining, no new data, and no change in latency.
- **The model was never the problem.** Tags were shipping at precision the model did not require: `gender_and_sex` at 0.2689 when 0.9894 was reachable at the same recall, `geolocation` at 0.0173 against 0.9655, `religion` at 0.0215 against 0.8278. The selector was choosing those points and recording them as successes.
- **The cost is recall**, and it is real: macro recall 0.6773 → 0.5668, worst priority recall 0.6267 → 0.4830. In a domain where a missed identifier is a reportable incident, that is a concession, and it is the reason both arms were built rather than one.
- **Keeping recall at 0.90 costs surprisingly little here:** `p80r90` scores *higher* micro F1 (0.8474) than `p80r70` at slightly lower precision. On sealed data the stricter arm is not the worse arm.
- **Still blocked**, on two corpora whose gold cannot support the target.
- **Precision had to be made gateable first** — the evaluator emits only recall per tag × corpus, so an 80% precision gate would have been NOT_MEASURABLE everywhere and blocked every arm while measuring none of them.

Date	2026-08-26
Project	projects/pii-target-8070
Run	26-08-26-1
Request	"try to get me 80% precision across the board and 70% recall"
Scope	3 arms × 8 sealed corpora, 61 labels; threshold re-selection only
CPU budget	1 core for latency; latency carried from the source model, see below
Outcome	large precision gain, both arms blocked , nothing promoted

Model scoreboard

Model	Dataset	F1 @5	F1 @3	F1 @1	Macro F1	Docs/sec	ms/doc
cascade-scorecard61-baseline	10360_betterdataai_ner_silver_eval_10.36k	0.225	0.180	0.126	0.263	248.2	4.0
cascade-scorecard61-baseline	10626_ai4privacy_pii_masking_eval_10.63k	0.758	0.711	0.611	0.715	248.2	4.0
cascade-scorecard61-baseline	20000_pii_holdout_20.00k	0.955	0.947	0.922	0.739	248.2	4.0
cascade-scorecard61-baseline	30000_pii2_eval_25.15k	0.926	0.911	0.862	0.761	248.2	4.0
cascade-scorecard61-baseline	38937_openpii_pii_eval_38.94k	0.992	0.988	0.979	0.692	248.2	4.0
cascade-scorecard61-baseline	4000_datax-dualjudge-evalset-1.32k	0.552	0.463	0.311		248.2	4.0
cascade-scorecard61-baseline	5617_nemotron_eval_5.36k					248.2	4.0
cascade-scorecard61-baseline	6589_govdocs2-dualjudge-eval20-3.53k	0.578	0.542	0.471		248.2	4.0
cascade-p80r70	10360_betterdataai_ner_silver_eval_10.36k	0.225	0.180	0.126	0.259	248.2	4.0
cascade-p80r70	10626_ai4privacy_pii_masking_eval_10.63k	0.758	0.711	0.611	0.732	248.2	4.0
cascade-p80r70	20000_pii_holdout_20.00k	0.955	0.947	0.923	0.827	248.2	4.0
cascade-p80r70	30000_pii2_eval_25.15k	0.926	0.911	0.862	0.809	248.2	4.0

Model	Dataset	F1 @5	F1 @3	F1 @1	Macro F1	Docs/ sec	ms/ doc
cascade-p80r70	38937_openpii_pii_eval_38.94k	0.99 2	0.98 8	0.97 9	0.717	248.2	4.0
cascade-p80r70	4000_datax-dualjudge-evalset-1.32k	0.55 2	0.46 2	0.31 1		248.2	4.0
cascade-p80r70	5617_nemotron_eval_5.36k					248.2	4.0
cascade-p80r70	6589_govdocs2-dualjudge-eval20-3.53k	0.57 8	0.54 2	0.47 1		248.2	4.0
cascade-p80r90	10360_betterdataai_ner_silver_eval_10.36k	0.22 5	0.18 0	0.12 6	0.268	248.2	4.0
cascade-p80r90	10626_ai4privacy_pii_masking_eval_10.63k	0.75 8	0.71 1	0.61 1	0.752	248.2	4.0
cascade-p80r90	20000_pii_holdout_20.00k	0.95 5	0.94 7	0.92 3	0.827	248.2	4.0
cascade-p80r90	30000_pii2_eval_25.15k	0.92 6	0.91 1	0.86 2	0.807	248.2	4.0
cascade-p80r90	38937_openpii_pii_eval_38.94k	0.99 2	0.98 8	0.97 9	0.725	248.2	4.0
cascade-p80r90	4000_datax-dualjudge-evalset-1.32k	0.55 2	0.46 2	0.31 1		248.2	4.0
cascade-p80r90	5617_nemotron_eval_5.36k					248.2	4.0
cascade-p80r90	6589_govdocs2-dualjudge-eval20-3.53k	0.57 8	0.54 2	0.47 1		248.2	4.0

Measured 2026-08-26 · n=10,360, n=10,626, n=20,000, n=30,000, n=38,937, n=4,000, n=5,617, n=6,589.